CONVEX OPTIMIZATION

- We are interested in a class of optimization problems where we can guarantee global optimality.
- When $f()$, the objective function, is a convex function and $g()$ and $h()$ are convex functions, we have a convex optimization problem.
- In this setting we have strong duality - the optimal solution of the primal problem is equal to the optimal solution of the dual problem.
- What is a convex function?

CONVEX FUNCTION

- First we need to know what is a convex set. A set C is a convex set if for any $x, y \in C$, $\theta x + (1 - \theta)y \in C$.
- For any two points lying in the convex set, a line joining them lies entirely in the convex set.
- Let a function $f : \mathbb{R}^d \to R$ be a function whose domain is a convex set C.
- The function is a convex function if for any $\boldsymbol{x}, \boldsymbol{y} \in C$, $f(\theta \boldsymbol{x} + (1 - \theta)\boldsymbol{y}) \leq \theta f(\boldsymbol{x}) + (1 - \theta) f(\boldsymbol{y})$
- Another way of looking at a convex function is to use the gradient: for any two points $\boldsymbol{x}$ and $\boldsymbol{y}$, we have $f(\boldsymbol{y}) \geq f(\boldsymbol{x}) + \nabla_{\boldsymbol{x}} f(\boldsymbol{x})(\boldsymbol{y} - \boldsymbol{x})$.

LINEAR PROGRAMMING

- Let us look at a convex optimization problem where the objective function and constraints are all linear.
- Such a convex optimization problem is called a linear programming problem.
- We can express a linear programming problem as $\min_{\boldsymbol{x}} \boldsymbol{c}^T\boldsymbol{x}$ subject to $\boldsymbol{Ax} \leq \boldsymbol{b}$ where $\boldsymbol{A} \in \mathbb{R}^{m\times d}$ and $b \in \mathbb{R}^{m\times 1}$.
- The Lagrangian $\mathfrak{L}(\boldsymbol{x}, \boldsymbol{\lambda})$ is given by $\mathfrak{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = c^T\boldsymbol{x} + \boldsymbol{\lambda}^T(\boldsymbol{Ax} - \boldsymbol{b})$ where $\boldsymbol{\lambda} \in \mathbb{R}^m$ is the vector of non-negative Lagrangian multipliers.
- We can rewrite the Lagrangian as $\mathfrak{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = (\boldsymbol{c} + \boldsymbol{A}^T\boldsymbol{\lambda})^T\boldsymbol{x} - \boldsymbol{\lambda}^T\boldsymbol{b}$.

LINEAR PROGRAMMING

- Taking the derivative of the Lagrangian with respect to $\boldsymbol{x}$ and setting it to zero we get $\boldsymbol{c} + \boldsymbol{A}^T\boldsymbol{\lambda} = 0$.
- Since $\mathfrak{D}(\lambda) = \min_{\boldsymbol{x}\in\mathbb{R}^d} \mathfrak{L}(\boldsymbol{x}, \boldsymbol{\lambda})$, plugging in the above equation gives $\mathfrak{D}(\boldsymbol{\lambda}) = -\boldsymbol{\lambda}^T\boldsymbol{b}$.
- We would like to maximize $\mathfrak{D}(\boldsymbol{\lambda})$, subject to the constraint $\boldsymbol{\lambda} \geq 0$.
- Thus we end up with the following problem:

$$\begin{aligned} max_{\boldsymbol{\lambda}\in\mathbb{R}^m} - \boldsymbol{\lambda}^T\boldsymbol{b} \\ \text{subject to } \boldsymbol{c} + \boldsymbol{A}^T\boldsymbol{\lambda} = 0 \\ \boldsymbol{\lambda} \geq 0 \end{aligned}$$

LINEAR PROGRAMMING

- We can solve the original primal linear program or the dual one - the optimum in each case is the same.
- The primal linear program is in *d* variables but the dual is in *m* variables, where *m* is the number of constraints in the original primal program.
- We choose to solve the primal or dual based on which of *m* or *d* is smaller.

QUADRATIC PROGRAMMING

- We now consider the case of a quadratic objective function subject to affine constraints:

$$min_{\mathbf{x}\in\mathbb{R}^d}\frac{1}{2}\mathbf{x}^T\mathbf{Q}\mathbf{x}+\mathbf{c}^T\mathbf{x} \text{ subject to}$$
$$\mathbf{A}\mathbf{x}\leq\mathbf{b}$$

- Here $\mathbf{A}\in\mathbb{R}^{m\times d}, b\in\mathbb{R}^m, c\in\mathbb{R}^d$

QUADRATIC PROGRAMMING

- The Lagrangian $\mathfrak{L}(\boldsymbol{x}, \boldsymbol{\lambda})$ is given by $\frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} + \boldsymbol{c}^T\boldsymbol{x} + \boldsymbol{\lambda}^T(\boldsymbol{A}\boldsymbol{x} - \boldsymbol{b})$.
- Rearranging the above we have $\mathfrak{L}(\boldsymbol{x}, \boldsymbol{\lambda}) = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} + (\boldsymbol{c} + \boldsymbol{A}^T\boldsymbol{\lambda})^T\boldsymbol{x} - \boldsymbol{\lambda}^T\boldsymbol{b}$
- Taking the derivative of $\mathfrak{L}(\boldsymbol{x}, \boldsymbol{\lambda})$ and setting it equal to zero gives $\boldsymbol{Q}\boldsymbol{x} + (\boldsymbol{c} + \boldsymbol{A}^T\boldsymbol{\lambda}) = 0$.
- If we take Q to be invertible, we have $\boldsymbol{x} = -\boldsymbol{Q}^{-1}(\boldsymbol{c} + \boldsymbol{A}^T\boldsymbol{\lambda})$.
- Plugging this value of $\boldsymbol{x}$ into $\mathfrak{L}(\boldsymbol{x}, \boldsymbol{\lambda})$ gives us $\mathfrak{D}(\boldsymbol{\lambda}) = -\frac{1}{2}(\boldsymbol{c} + \boldsymbol{A}^T\boldsymbol{\lambda})^T\boldsymbol{Q}^{-1}(\boldsymbol{c} + \boldsymbol{A}^T\boldsymbol{\lambda}) - \boldsymbol{\lambda}^T\boldsymbol{b}$.
- This gives us the dual optimization problem: $\max_{\boldsymbol{\lambda} \in \mathbb{R}^m} -\frac{1}{2}(\boldsymbol{c} + \boldsymbol{A}^T\boldsymbol{\lambda})^T\boldsymbol{Q}^{-1}(\boldsymbol{c} + \boldsymbol{A}^T\boldsymbol{\lambda}) - \boldsymbol{\lambda}^T\boldsymbol{b}$ subject to $\boldsymbol{\lambda} \geq \boldsymbol{0}$.

LAGRANGIAN FORMULATION

- In contrast to the original formulation $\mathfrak{D}(\boldsymbol{\lambda}) = \min_{\boldsymbol{x} \in \mathbb{R}^d} \mathfrak{L}(\boldsymbol{x}, \boldsymbol{\lambda})$ is an unconstrained optimization problem for a given value of $\boldsymbol{\lambda}$.
- We observe that $\mathfrak{D}(\boldsymbol{\lambda}) = \min_{\boldsymbol{x} \in \mathbb{R}^d} \mathfrak{L}(\boldsymbol{x}, \boldsymbol{\lambda})$ is a point-wise minimum of affine functions and hence $\mathfrak{D}(\boldsymbol{\lambda})$ is concave even though $f()$ and $g()$ may be nonconvex.
- We have obtained a Lagrangian formulation for a constrained optimization problem where the constraints are inequalities. What happens when some constraints are equalities?

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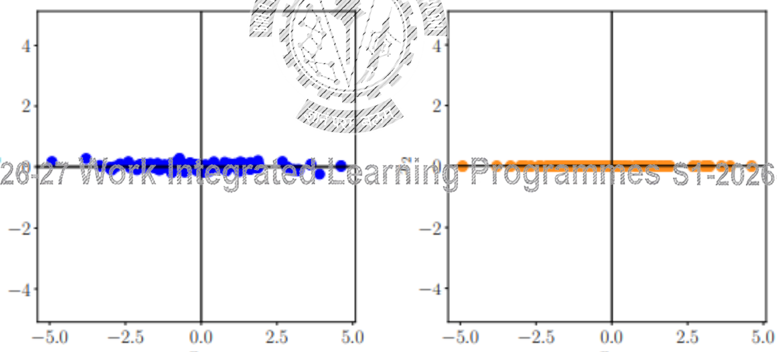
Topic: # 12 Principle Components Analysis

INTRODUCTION

- We will look at principle components analysis and dimension reduction in this lecture.
- High-dimensional data is hard to visulaize and interpret, can we project this data into lower dimensions while preserving the semantics of the data so as to draw the same conclusions as if we interpreted the higher dimensional data?
- Higher dimensional data is often overcomplete, in that there are redundant dimensions which can be explained by a combination of other dimensions.
- Dimensions in higher-dimensional data might be correlated, so the actual data may have an intrinsic lower-dimensional structure

PRINCIPLE COMPONENTS ANALYSIS

- PCA is a technique for linear dimensionality reduction. It was first proposed by Pearson in 1900 and was independently rediscovered by Hotelling in 1933.



(a) Dataset with x_1 and x_2 coordinates.

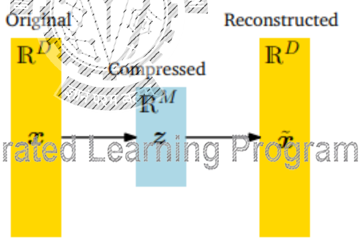
(b) Compressed dataset where only the x_1 coordinate is relevant.

PROBLEM SETTING

- **Given :** Let $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n\}$, $\boldsymbol{x}_n \in \mathbb{R}^D$ an independent, identically distributed dataset, with mean $\mathbf{0}$.
- Thus, the data covariance matrix $\boldsymbol{S} = \frac{1}{N}\sum_{n=1}^{n=N} \boldsymbol{x_n}\boldsymbol{x_n}^T$.
- **Aim :** To find projections $\tilde{\boldsymbol{x}}_n \in U \subseteq \mathbb{R}^D$ of datapoints $\boldsymbol{x}_n \in \mathbb{R}^D$ which are as similar as possible to the original datapoints but $\dim(U) = M < D$.
- We are looking for a lower-dimensional compressed representation $\boldsymbol{z}_n$ of $\boldsymbol{x}_n$ such that $\boldsymbol{z}_n = \boldsymbol{B}^T\boldsymbol{x}_n$ where the projection matrix $\boldsymbol{B} = [\boldsymbol{b}_1, \ldots, \boldsymbol{b}_m] \in \mathbb{R}^{D\times M}$.
- The columns of B are orthonormal which means $\boldsymbol{b}_i^T\boldsymbol{b}_j = 0$ when $i \neq j$ and $\boldsymbol{b}_i^T\boldsymbol{b}_i = 1$.

PROBLEM SETTING

- The figure below shows how $\boldsymbol{z}$ represents the lower-dimensional representation of the compressed data $\tilde{\boldsymbol{x}}$ and plays the role of a bottleneck which controls the information flow between $\boldsymbol{x}$ and $\tilde{\boldsymbol{x}}$.



- There exists a linear relationship between the original data $\boldsymbol{x}$, its low-dimensional code $\boldsymbol{z}$ and the compressed data $\tilde{\boldsymbol{x}}$: $\boldsymbol{z} = \boldsymbol{B}^T\boldsymbol{x}$, and $\tilde{\boldsymbol{x}} = \boldsymbol{B}\boldsymbol{z}$ for a suitable matrix $\boldsymbol{B}$.

MAXIMUM VARIANCE PERSPECTIVE

- We can interpret information content in the data as how ”space-filling” it is and describe the information contained in the data by looking at the spread of the data.
- We can capture spread of the data using the concept of variance.
- PCA can then be viewed as a dimensionality reduction algorithm that maximizes the variance in the low-dimensional representation of the data to retain as much information as possible.
- Mathematically our aim is to find a matrix $\boldsymbol{B}$ so that we can retain as much information as possible by projecting the data on the columns $\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots \boldsymbol{b}_M$ of the matrix.

CENTRED DATA

- If $\boldsymbol{\mu}$ is the mean of the data. Centred data means that we work with data columns $\boldsymbol{x} - \boldsymbol{\mu}$, rather than the original columns $\boldsymbol{x}$.
- Note that $\mathbb{V}_{\boldsymbol{z}}(\boldsymbol{z}) = \mathbb{V}_{\boldsymbol{x}}(\boldsymbol{B}^T(\boldsymbol{x} - \boldsymbol{\mu})) = \mathbb{V}_{\boldsymbol{x}}(\boldsymbol{B}^T\boldsymbol{x} - \boldsymbol{B}^T\boldsymbol{\mu}) = \mathbb{V}_{\boldsymbol{x}}(\boldsymbol{B}^T\boldsymbol{x})$.
- Thus by considering $\boldsymbol{x} - \boldsymbol{\mu}$, the variance does not change. Therefore we assume that the data has a mean of $\boldsymbol{0}$ for this lecture.
- Letting the mean be $\mathbb{E}_{\boldsymbol{x}}(\boldsymbol{x}) = \boldsymbol{0}$ means $\mathbb{E}_{\boldsymbol{z}}(\boldsymbol{z}) = \mathbb{E}_{\boldsymbol{x}}(\boldsymbol{B}^T\boldsymbol{x}) = \boldsymbol{B}^T\mathbb{E}_{\boldsymbol{x}}(\boldsymbol{x}) = \boldsymbol{0}$
- And the data covariance matrix $\boldsymbol{S} = \frac{1}{N}\sum_{n=1}^{n=N} \boldsymbol{x_n}\boldsymbol{x_n}^T$.

DIRECTION WITH MAXIMAL VARIANCE

- We maximize the variance of the low-dimensional code by following a sequential approach.
- **Aim 1 :** To maximize the variance V_1 of the first coordinate z_{1n} of $\boldsymbol{z} \in R^M$
- i.e to maximize $V_1 = \mathbb{V}(z_1) = \frac{1}{N}\sum_{n=1}^{n=N} z_{1n}^2$ since the data $\boldsymbol{x}$ is independent.
- Now $z_{1n} = \boldsymbol{b}_1^T\boldsymbol{x}_n$, and can be viewed as the orthogonal projection of $\boldsymbol{x}_n$ onto the one-dimensional subspace spanned by $\boldsymbol{b}_1$.

DIRECTION WITH MAXIMAL VARIANCE

$$\begin{aligned} \text{Then we have } V_1 &= \frac{1}{N}\sum_{n=1}^{n=N}(\boldsymbol{b}_1^T\boldsymbol{x}_n)^2 \\ &= \frac{1}{N}\sum_{n=1}^{n=N}\boldsymbol{b}_1^T\boldsymbol{x}_n\boldsymbol{x}_n^T\boldsymbol{b}_1 \\ &= \boldsymbol{b}_1^T\Big(\sum_{n=1}^{n=N}\boldsymbol{x}_n\boldsymbol{x}_n^T\Big)\boldsymbol{b}_1 \\ &= \boldsymbol{b}_1^T\boldsymbol{S}\boldsymbol{b}_1. \end{aligned}$$

Arbitrarily increasing the magnitude of the vector $\boldsymbol{b}_1$ will increase the variance - so we seek to maximize the variance subject to $\|\boldsymbol{b}_1\| = 1$.

DIRECTION WITH MAXIMAL VARIANCE

- Therefore to find the direction $\boldsymbol{b}_1$ that maximizes variance can be set up as a constrained optimization problem

$$\max\ \boldsymbol{b}_1^T\boldsymbol{S}\boldsymbol{b}_1 \text{ subject to } \|\boldsymbol{b}_1\| = 1$$

- To solve this problem we set up the Lagrangian $\mathfrak{L}(\boldsymbol{x}, \lambda) = \boldsymbol{b}_1^T\boldsymbol{S}\boldsymbol{b}_1 + \lambda(1 - \boldsymbol{b}_1^T\boldsymbol{b}_1)$.

SOLVING THE LAGRANGIAN

- To solve the Lagrangian, let $\frac{\partial \mathbb{L}}{\partial \lambda} = 0$ and $\frac{\partial \mathbb{L}}{\partial \boldsymbol{b}_1} = \mathbf{0}$.
- So, we get

$$\frac{\partial \mathbb{L}}{\partial \lambda} = 1 - \boldsymbol{b}_1^T \boldsymbol{b}_1 = 0$$

$$\frac{\partial \mathbb{L}}{\partial \boldsymbol{b}_1} = 2\boldsymbol{b}_1^T \boldsymbol{S} - 2\lambda \boldsymbol{b}_1^T = 0$$

- On simplification, we have $\boldsymbol{S}\boldsymbol{b}_1 = \lambda \boldsymbol{b}_1$ and $\boldsymbol{b}_1^T \boldsymbol{b} = 1$.
- Thus we find that the direction $\boldsymbol{b}_1$ we seek is an eigenvector of the covariance matrix $\boldsymbol{S}$ and λ is its corresponding eigenvalue.

FIRST PRINCIPAL COMPONENT

- Putting the result of the previous slide into the objective function of the constrained optimization problem ie. max $\boldsymbol{b}_1^T \boldsymbol{S}\boldsymbol{b}_1$ we have $\boldsymbol{b}_1^T \boldsymbol{S}\boldsymbol{b}_1 = \boldsymbol{b}_1^T \lambda \boldsymbol{b}_1 = \lambda$.
- Our objective function boils to maximizing λ which means we are looking for the eigenvector of $\boldsymbol{S}$ that corresponds to its largest eigenvalue.
- This is the first principal component.
- Let us now examine the inner workings of the Lagrangian method.

WHY DOES THIS METHOD WORK?

- Suppose we have the following constrained optimization problem:

$$\max f(x, y) \text{ subject to}$$
$$g(x, y) = c$$

- We note that at the optimal solution (x_0, y_0), if we move a small distance $(\delta x, \delta y)$, we must continue to remain on the surface $g(x, y) = c$. This means $g(x_0 + \delta x, y_0 + \delta y) = c = g(x, y)$.
- But $dg = g(x_0 + \delta x, y_0 + \delta y) - g(x, y) = \nabla g.(\delta x, \delta y) = 0$.
- Therefore ∇g is orthogonal to the displacement vector $(\delta x, \delta y)$.

WHY DOES THIS METHOD WORK?

- As we move along the displacement vector $(\delta x, \delta y)$, the value of the objective function also cannot change because otherwise we can get a better solution by moving along the displacement vector or its negative direction.
- Thus we have $\nabla f.(\delta x, \delta y) = 0$, so ∇f is orthogonal to the displacement vector $(\delta x, \delta y)$.
- Therefore, ∇f and ∇g are orthogonal to the displacement vector $(\delta x, \delta y)$ and hence $\nabla f = \lambda \nabla g$
- This leads us to $\frac{\partial \mathbb{L}}{\partial \boldsymbol{b}_1} = 0$.
- $\frac{\partial \mathbb{L}}{\partial \lambda} = 0$ merely enforces the constraint in the original constrained optimization problem.

M-DIMENSIONAL SUBSPACE WITH MAXIMUM VARIANCE

- Assume that we have found the first $m-1$ principal components as the $m-1$ eigenvectors of S that are associated with the largest $m-1$ eigenvalues of S.
- Since S is symmetric we can use the spectral theorem to use the $m-1$ eigenvectors to construct an orthonormal basis of an $m-1$-dimensional basis of $\mathbb{R}^D$.
- The *m*th principal component can be found by subtracting from the data the contribution of the first $m-1$ components $\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots \boldsymbol{b}_{m-1}$. Essentially we are trying to find principal components that compress the remainder of the information.

M-DIMENSIONAL SUBSPACE WITH MAXIMUM VARIANCE

- $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N] \in \mathbb{R}^{D\times N}$ contains the data points $\boldsymbol{x}_k$ as column vectors.
- Then, new data matrix $\hat{\boldsymbol{X}}$ is given by

$$\begin{aligned}\hat{\boldsymbol{X}} &= \boldsymbol{X} - \sum_{i=1}^{m-1} \boldsymbol{b}_i \boldsymbol{b}_i^T \boldsymbol{X} \\ &= \boldsymbol{X} - \boldsymbol{B}_{m-1}\boldsymbol{X}\end{aligned}$$

- Here $\boldsymbol{B}_{m-1} = \sum_{i=1}^{m-1} \boldsymbol{b}_i \boldsymbol{b}_i^T$ is a projection matrix that projects $\boldsymbol{X}$ onto the subspace spanned by $\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots \boldsymbol{b}_{m-1}$.

M-DIMENSIONAL SUBSPACE WITH MAXIMUM VARIANCE

- To find the *m*th principal component we maximize

$$\begin{aligned} V_m = \mathbb{V}[z_m] &= \frac{1}{N}\sum_{n=1}^{n=N} z_{mn}^2 \\ &= \frac{1}{N}\sum_{n=1}^{n=N} (\boldsymbol{b}_m^T \boldsymbol{x}_n)^2 \\ &= \boldsymbol{b}_m^T \hat{\boldsymbol{S}} \boldsymbol{b}_m. \end{aligned}$$

- Here $\hat{\boldsymbol{S}}$ is the data covariance matrix of the transformed data set represented by $\hat{\boldsymbol{x}} = [\hat{\boldsymbol{x}}_1, \hat{\boldsymbol{x}}_2, \ldots \hat{\boldsymbol{x}}_N]$.

M-DIMENSIONAL SUBSPACE WITH MAXIMUM VARIANCE

- As before we set up a constrained optimization problem to find the first principal component, and establish that the optimal solution $\boldsymbol{b}_m$ is the eignevector of $\hat{\boldsymbol{S}}$ that corresponds to the largest eigenvalue.
- We now establish that $\boldsymbol{b}_m$ is an eigenvector of the original data matrix $\boldsymbol{X}$.
- More generally the sets of eigenvectors for $\hat{\boldsymbol{S}}$ and $\boldsymbol{S}$ are the same.

EIGENVECTORS OF $\boldsymbol{S}$ AND $\hat{\boldsymbol{S}}$

- We now show that the eigenvectors of $\boldsymbol{S}$ and $\hat{\boldsymbol{S}}$ are the same.
- Let $\boldsymbol{b}_i$ be an eigenvector of $\boldsymbol{S}$, i.e $\boldsymbol{S}\boldsymbol{b}_i = \lambda \boldsymbol{b}_i$.
- Now we can write

$$\begin{aligned}
\hat{\boldsymbol{S}}\boldsymbol{b}_i &= \frac{1}{N}(\boldsymbol{X} - \boldsymbol{B}_{m-1}\boldsymbol{X})(\boldsymbol{X} - \boldsymbol{B}_{m-1}\boldsymbol{X})^T\boldsymbol{b}_i \\
&= (\boldsymbol{S} - \boldsymbol{S}\boldsymbol{B}_{m-1}^T - \boldsymbol{B}_{m-1}\boldsymbol{S} + \boldsymbol{B}_{m-1}\boldsymbol{S}\boldsymbol{B}_{m-1}^T)\boldsymbol{b}_i \\
&= (\boldsymbol{S} - \boldsymbol{S}\boldsymbol{B}_{m-1} - \boldsymbol{B}_{m-1}\boldsymbol{S} + \boldsymbol{B}_{m-1}\boldsymbol{S}\boldsymbol{B}_{m-1})\boldsymbol{b}_i
\end{aligned}$$

- Note that in the last line we have used the fact that $\boldsymbol{B}_{m-1}$ is a projection matrix and is therefore symmetric.

EIGENVECTORS OF $\boldsymbol{S}$ AND $\hat{\boldsymbol{S}}$

- **Case 1:** $i \geq m$.
- $\boldsymbol{b}_i$ is an eigenvector not among the first $m-1$ components.
- Since $\boldsymbol{B}_{m-1} = \sum_{i=1}^{m-1} \boldsymbol{b}_i\boldsymbol{b}_i^T$ and $\boldsymbol{b}_m$ is orthogonal to the $\boldsymbol{b}_i$, $1 \leq i \leq m-1$, we have $\boldsymbol{B}_{m-1}\boldsymbol{b}_i = 0$.
- Plugging this into the last equation on the previous slide, we see that $\hat{\boldsymbol{S}}\boldsymbol{b}_i = (\boldsymbol{S} - \boldsymbol{B}_{m-1}\boldsymbol{S})\boldsymbol{b}_i = \boldsymbol{S}\boldsymbol{b}_i = \lambda_i\boldsymbol{b}_i$.
- Thus $\boldsymbol{S}\boldsymbol{b}_m = \lambda_m\boldsymbol{b}_m$. λ_m is the mth largest eigenvalue of $\boldsymbol{S}$ and is also the largest eigenvalue of $\hat{\boldsymbol{S}}$ because of the way the constrained optimization problem is set up.

EIGENVECTORS OF $\boldsymbol{S}$ AND $\hat{\boldsymbol{S}}$

- **Case 2:** $i \leq m - 1$.
- We have $\boldsymbol{B}_{m-1}\boldsymbol{b}_i = \sum_{j=1}^{m-1} \boldsymbol{b}_j\boldsymbol{b}_j^T\boldsymbol{b}_i = \boldsymbol{b}_i$.
- Plugging this into $\hat{\boldsymbol{S}}\boldsymbol{b}_i = (\boldsymbol{S} - \boldsymbol{S}\boldsymbol{B}_{m-1} - \boldsymbol{B}_{m-1}\boldsymbol{S} + \boldsymbol{B}_{m-1}\boldsymbol{S}\boldsymbol{B}_{m-1})\boldsymbol{b}_i$, we get $\boldsymbol{S}\boldsymbol{b}_i = \boldsymbol{0}$.
- Thus the vectors $\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots \boldsymbol{b}_{m-1}$ are eigenvectors for $\hat{\boldsymbol{S}}$ which are associated with the eigenvalue 0.

- Since $V_m = \boldsymbol{b}_m^T\boldsymbol{S}\boldsymbol{b}_m = \lambda_m$, we see that the variance of the data projected onto the mth principal component is λ_m.
- To find an M-dimensional susbspace that retains as much information as possible, PCA tells us to choose the columns of the matrix $\boldsymbol{B}$ as the M eigenvectors of the data covariance matrix $\boldsymbol{S}$ that have the largest eigenvalues.

PROJECTION PERSPECTIVE

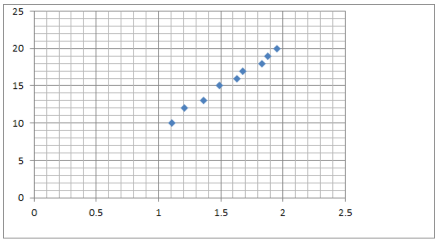
- We derived the PCA as an algorithm that maximizes the variance in the projected space to retain as much information as possible.
- Now we can also derive the PCA using a projection perspective to minimize the average reconstruction error. The original data is modeled as $\boldsymbol{x}_n$ and the reconstruction is modeled as $\tilde{\boldsymbol{x}}_n$. We seek to minimize the distance between $\boldsymbol{x}_n$ and $\boldsymbol{x}_n$.

AN EXAMPLE

Consider the data given below:

x1	x2	x3	x4	x5	x6	x7	x8	x9	Me
1.11	1.21	1.36	1.49	1.63	1.68	1.83	1.88	1.95	1.5711
10	12	13	15	16	17	18	19	20	15.555

Data when plotted, we get



After subtracting mean from the data, then we get

$$
\begin{aligned}
S &= \frac{1}{9}XX' \\
&= \begin{pmatrix} 0.079498765 & 0.888271605 \\ 0.888271605 & 10.02469136 \end{pmatrix}
\end{aligned}
$$

The largest eigenvalue of S is 10.103 and the corresponding eigenvector with unit norm (first principal component) is $b_1 = [0.088269, 0.996097]^T$.

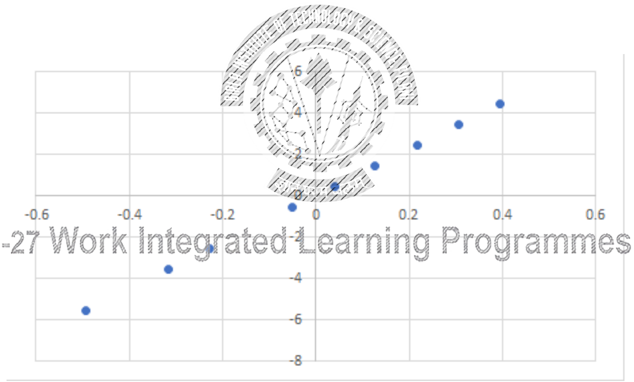
The compressed or the reduced data is then given by $z = b_1^T X$

z1	z2	z3	z4	z5	z6	z7	z8	z9
-5.57	-3.57	-2.56	-0.56	0.45	1.45	2.46	3.46	4.46

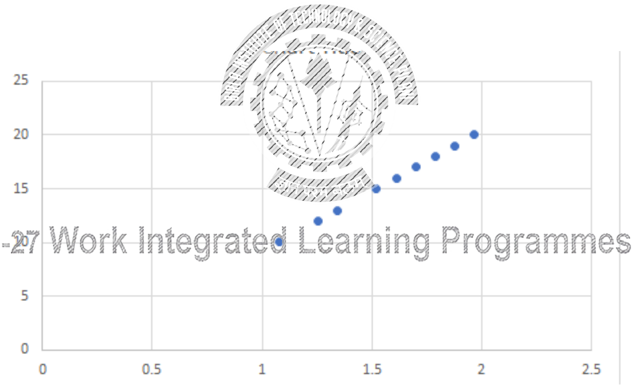
We can project the data onto the principal subspace by $\tilde{X} = b_1 z$. To obtain our projection in the original data space (i.e., before standardization), we need to undo the standardization.

$$\tilde{X} = \tilde{X} + \mu$$

PROJECTED DATA ONTO THE PRINCIPAL SUBSPACE



PROJECTED DATA ONTO THE PRINCIPAL SUBSPACE IN THE ORIGINAL DATA SPACE



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Topic: # 13 Principle Components Analysis

INTRODUCTION

- In the previous lecture, we discussed dimensionality reduction using PCA
- In this lecture, we will see the use of linear algebra in practical implmentation of PCA.
- We will also study the challenges encountered when PCA is used in problems of larger dimensions.
- Finally, we elaborate the key steps of PCA in practice.

REVISION OF PCA PROBLEM

- We derived the matrix **B** used in generation of lower-dimensional representation $\boldsymbol{z}$ and the compressed data $\tilde{\boldsymbol{x}}$.
- The data covariance matrix **S** was used to derive **B**
- Recall that linear relationship connecting the original data $\boldsymbol{x}$, its low-dimensional code $\boldsymbol{z}$ and the compressed data $\tilde{\boldsymbol{x}}$: $\boldsymbol{z} = \boldsymbol{B}^T\boldsymbol{x}$, and $\tilde{\boldsymbol{x}} = \boldsymbol{B}\boldsymbol{z}$.

EIGENVECTOR COMPUTATION AND LOW RANK APPROXIMATION

- In the previous sections, we obtained the basis of the principal subspace as the eigenvectors that are associated with the largest eigenvalues of the data covariance matrix.

$$S = \frac{1}{N}\sum_{i=1}^{N} x_n x_n^T \tag{1}$$

- Equivalently we get

$$S = \frac{1}{N}XX^T \tag{2}$$

$$X = [x_1, ..., x_N] \in \mathbb{R}^{D\times N} \tag{3}$$

- Note that X is a $D \times N$ matrix,

EIGENVECTOR COMPUTATION AND LOW RANK APPROXIMATION

- To get the eigenvalues and the corresponding eigenvectors of S, we can follow two approaches
- We can perform an eigendecomposition and compute the eigenvalues and eigenvectors of S directly.
- We can also use a singular value decomposition. Since S is symmetric and factorizes into XX^T, the eigenvalues of S are the squared singular values of X.
- Assume the SVD of X as $X = U\Sigma V^T$. Then

$$S = \frac{1}{N}XX^T = \frac{1}{N}U\Sigma\Sigma^T U^T$$

EIGENVECTOR COMPUTATION AND LOW RANK APPROXIMATION

- The columns of U are the eigenvectors of S.
- The eigenvalues λ_d of S are related to the singular values of X via

$$\lambda_d = \frac{\sigma_d^2}{N} \tag{4}$$

- This relationship between the eigenvalues of S and the singular values of X provides the connection between the maximum variance view and the singular value decomposition.

PCA USING LOW-RANK MATRIX APPROXIMATIONS

- To maximize the variance of the projected data PCA chooses the columns of U to be the eigenvectors that are associated with the M largest eigenvalues of the data covariance matrix S
- The Eckart-Young theoremoffers a direct way to estimate the low-dimensional representation.
- Consider the best rank-M approximation of X defined as $\tilde{X}_M$

$$\tilde{X}_M = \text{argmin}_{rank(A)<=M} \|X - A\|_2 \qquad (5)$$

PCA USING LOW-RANK MATRIX APPROXIMATIONS

- The Eckart-Young theorem states that the best rank M approximation $\tilde{X}_M$ is given by truncating the SVD at the top-M singular value.

$$\tilde{X}_M = U_M \Sigma_M V_M^T$$

- U_M is an orthogonal matrix
- V_M is an orthogonal matrix
- Σ_M has M largest singular values of X as diagonal entries

PRACTICAL ASPECTS OF PCA

- Finding eigenvalues and eigenvectors is also important in other fundamental machine learning methods that require matrix decompositions
- In theory we can solve for the eigenvalues as roots of the characteristic polynomial.
- However for matrices larger than 4 by 4 this is not possible because we would need to find the roots of polynomial of degree 5 or higher.

PRACTICAL ASPECTS OF PCA

- However the Abel-Ruffini theorem states that there exists no algebraic solution to this problem for polynomials of degree 5 or more.
- Therefore, in practice, solve for eigenvalues or singular values using iterative methods, which are implemented in all modern packages for linear algebra
- In many applications we only require a few eigenvectors.

PRACTICAL ASPECTS OF PCA

- It would be wasteful to compute the full decomposition,and then discard all eigenvectors with eigenvalues that are beyond the first few.
- It turns out that if we are interested in only the first few eigenvectors (with the largest eigenvalues), then iterative processes, which directly optimize these eigenvectors, are computationally more efficient than a full eigendecomposition

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PRACTICAL ASPECTS OF PCA

- In the extreme case of only needing the first eigenvector, a simple method called the power iteration is very efficient.
- Power iteration chooses a random vector x_0 that is not in the null space of S and follows the iteration for $k = 0, 1, \ldots$

$$x_{k+1} = \frac{Sx_k}{||Sx_k||} \quad (6)$$

- This sequence of vectors converges to the eigenvector associated with the largest eigenvalue of S.

PCA IN HIGH DIMENSION

1. In order to do PCA, we need to compute the data covariance matrix.
2. In D dimensions, the data covariance matrix is a $D \times D$ matrix.
3. Computing the eigenvalues and eigenvectors of this matrix is computationally expensive as it scales cubically in D.
4. Therefore, PCA, as we discussed earlier, will be infeasible in very high dimensions.
5. In the following, we provide a solution to this problem for the case that we have substantially fewer data points than dimensions, i.e., $N << D$

PCA IN HIGH DIMENSION

- Assume we have a centered dataset $x_1, \ldots, x_N$, $x_n \in \mathbb{R}^D$. Then the data covariance matrix is given as $S = \frac{1}{N}XX^T$
-

$$Sb_m = \frac{1}{N}XX^T b_m = \lambda_m b_m \tag{7}$$

$$\frac{1}{N}X^T XX^T b_m = \lambda_m X^T b_m \tag{8}$$

$$\frac{1}{N}X^T X c_m = \lambda_m c_m \tag{9}$$

- Here $c_m = X^T b_m$.
- The nonzeroeigenvalues of XX^T is same as the nonzero eigenvalues of $X^T X$.

PCA IN HIGH DIMENSION

- Now that we have the eigenvectors of $\frac{1}{N}X^TX$,
- We need to derive the eigenvectors of XX^T, which we still need for PCA

$$\frac{1}{N}XX^TXc_m = \lambda_m Xc_m \tag{10}$$

- Here, we recover the data covariance matrix again.
- This now also means that we recover Xc_m as an eigenvector of S.

KEY STEPS OF PCA IN PRACTICE

- In the following, we will go through the individual steps of PCA using a running example.
- We are given a two dimensional dataset and we want to use PCA to project it onto a one-dimensional subspace. The key steps are given below
  - Mean subtraction
  - Standardization
  - Eigendecomposition of the covariance matrix
  - Projection

KEY STEPS OF PCA IN PRACTICE

Mean subtraction

1. We start by centering the data by computing the mean of the dataset and subtracting it from every single data point.
2. This ensures that the dataset has mean 0.
3. Mean subtraction is not strictly necessary but reduces the risk of numerical problems

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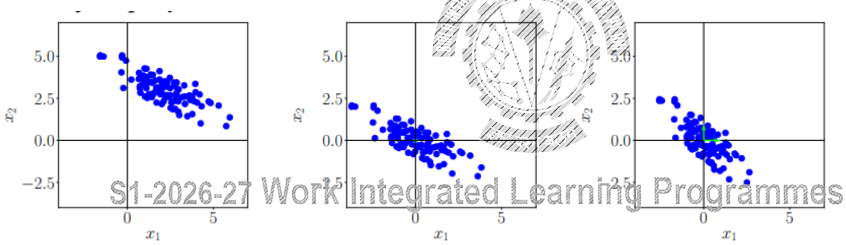
KEY STEPS OF PCA IN PRACTICE

Standardization

1. Divide the data points by the standard deviation σ_d of the dataset for every dimension d.
2. Now the data is unit free, and it has variance 1 along each axis, which is indicated by the standardization
3. This step completes the standardization of the data.

KEY STEPS OF PCA IN PRACTICE

Figure: PCA



(a) Original dataset.

(b) Step 1: Centering by subtracting the mean from each data point.

(c) Step 2: Dividing by the standard deviation to make the data unit free. Data has variance 1 along each axis.

Figure 10.11 Steps of PCA. (a) Original dataset; (b) centering; (c) divide by standard deviation; (d) eigendecomposition; (e) projection; (f) mapping back to original data space.

KEY STEPS OF PCA IN PRACTICE

Eigendecomposition of the covariance matrix

1. Compute the data covariance matrix and its eigenvalues and corresponding eigenvectors.
2. Since the covariance matrix is symmetric, the spectral theorem states that we can find an orthonormal basis of eigenvectors.
3. The eigenvectors are scaled by the magnitude of the corresponding eigenvalue.

KEY STEPS OF PCA IN PRACTICE

Projection

1. We can project any data point $x_* \in \mathbb{R}^d$ onto the principal subspace:
2. To get this right, we need to standardize x_* using the mean and standard deviation of the training data in the d th dimension

$$x_*^{(d)} = \frac{x_*^{(d)} - \mu_d}{\sigma_d}, \quad d = 1, ..., D \tag{11}$$

3. Here $x_*^{(d)}$ is the d th component of x_*.

KEY STEPS OF PCA IN PRACTICE

1. We obtain the projection as

$$\tilde{x} = BB^T x_* \tag{12}$$

2. The coordinates are

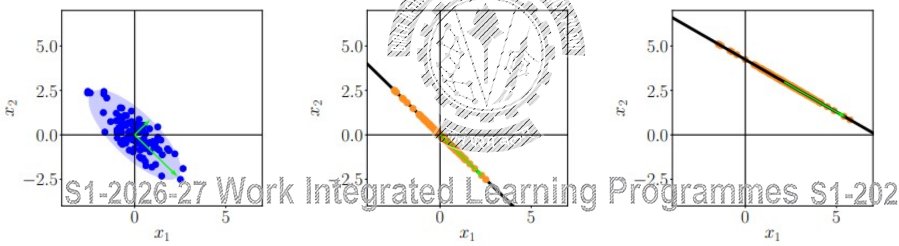
$$z_* = B^T x_* \tag{13}$$

with respect to the basis of the principal subspace.

3. Here, B is the matrix that contains the eigenvectors that are associated with the largest eigenvalues of the data covariance matrix as columns.

KEY STEPS OF PCA IN PRACTICE

Figure: PCA



(d) Step 3: Compute eigenvalues and eigenvectors (arrows) of the data covariance matrix (ellipse).

(e) Step 4: Project data onto the principal subspace.

(f) Undo the standardization and move projected data back into the original data space from (a).